

ORF 524 — Final Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 Mean Square Error Prediction (25 points)

Assume Y, W, X be random variables with $\mathbb{E}[Y^2] < \infty, \mathbb{E}[W^2] < \infty$. Let $\mathcal{G} = \{g : \mathbb{E}[|g(X)|^2] < \infty\}$.

1. Show that

$$\begin{aligned}\mathbb{E}[(Y - g_1(X))(W - g_2(X))] &= \mathbb{E}[(Y - \mathbb{E}[Y|X])(W - \mathbb{E}[W|X])] \\ &\quad + \mathbb{E}[(\mathbb{E}[Y|X] - g_1(X))(\mathbb{E}[W|X] - g_2(X))],\end{aligned}$$

for all $g_1, g_2 \in \mathcal{G}$.

2. Show that

$$\mathbb{E}[\mathbb{V}[Y|X]] = \min_{g \in \mathcal{G}} \mathbb{E}[(Y - g(X))^2].$$

3. Show that

$$\mathbb{V}[Y] = \mathbb{E}[\mathbb{V}[Y|X]] + \mathbb{V}[\mathbb{E}[Y|X]].$$

4. Show that

$$\mathbb{Cov}[Y, W] = \mathbb{E}[\mathbb{Cov}[Y, W|X]] + \mathbb{Cov}[\mathbb{E}[Y|X], \mathbb{E}[W|X]],$$

where $\mathbb{Cov}[Y, W|X] = \mathbb{E}[(Y - \mathbb{E}[Y|X])(W - \mathbb{E}[W|X])|X]$.

5. Show that

$$\mathbb{Cov}[Y, W] \leq \sqrt{\mathbb{E}[\mathbb{V}[Y|X]\mathbb{V}[W|X]]} + \sqrt{\mathbb{V}[\mathbb{E}[Y|X]]\mathbb{V}[\mathbb{E}[W|X]]}.$$

2 Conservative Inference via Markov Inequality (25 points)

Let $(X_i : 1 \leq i \leq n)$ be pairwise uncorrelated random variables with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}[X_i] = \sigma^2$ for all $i = 1, \dots, n$.

1. For each $i = 1, \dots, n$, and $g \in \mathcal{G}_i = \{g : \mathbb{E}[g(X_i)] < \infty, g \geq 0\}$, show that

$$\mathbb{E}[g(X_i)] \geq \mathbb{E}[g(X_i)\mathbb{1}(g(X_i) > \epsilon)] \geq \epsilon \mathbb{P}[g(X_i) > \epsilon], \quad \text{for all } \epsilon > 0.$$

2. Show that

$$\mathbb{P}[|\bar{X}_n - \mu| > \epsilon] \leq \frac{\sigma^2}{n\epsilon^2}.$$

3. Consider a Wald-type confidence interval for μ of the form

$$I = [\bar{X}_n - q, \bar{X}_n + q]$$

for some $q \in \mathbb{R}$. Let $\alpha \in [0, 1]$.

- (a) Find a q such that $\mathbb{P}[\mu \in I] \geq 1 - \alpha$.
- (b) Assume that $\{X_i\}$ are independent, distributed $X_i \sim \text{Normal}(\mu, \sigma^2)$, and find a q such that $\mathbb{P}[\mu \in I] = 1 - \alpha$.
- (c) Compute the interval length of the two confidence intervals constructed above. Which one is better? Discuss, in particular, their difference in light of the role of robustness vs. efficiency in statistical inference.

3 M-Estimation with Missing Data (50 points)

Let $(y_i, t_i, \mathbf{x}_i')$, $i = 1, \dots, n$, be an observed random sample, where $y_i = t_i y_i^*$ with y_i^* taking values in $\mathbb{R}$, t_i taking values $\{0, 1\}$, and $\mathbf{x}_i$ taking values in $\mathbb{R}^d$. The parameter of interest is

$$\theta_0 = \arg \min_{\theta \in \Theta} \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$$

where $\rho : \mathbb{R}^{d+1} \times \Theta \mapsto \mathbb{R}$ is a smooth loss function, and all necessary moments are assumed to exist.

1. Assume that $(y_i^*, \mathbf{x}_i) \perp\!\!\!\perp t_i$ (i.e., missing completely at random).

- (a) Show that $\mathbb{E}[t_i \rho(y_i, \mathbf{x}_i; \theta)]/p = \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where $p = \mathbb{P}(t_i = 1) > 0$.
- (b) Provide sufficient conditions for $Q_n(\theta) \rightarrow_{\mathbb{P}} \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where

$$Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{\hat{p}_n} \quad \text{and} \quad \hat{p}_n = \frac{1}{n} \sum_{i=1}^n t_i.$$

(c) The previous results motivate the estimator

$$\tilde{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{\hat{p}_n},$$

which is consistent for θ_0 under regularity conditions on $\theta \mapsto \rho(\cdot, \cdot; \theta)$.

Provide sufficient conditions so that

$$\sqrt{n}(\tilde{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_i + o_{\mathbb{P}}(1) \rightsquigarrow \text{Normal}(0, \Omega_0),$$

where $(\varphi_i : i = 1, \dots, n)$ are mean-zero i.i.d., and identify the exact form of φ_i and Ω_0 .

- (d) Propose a consistent estimator of Ω_0 , and show it is consistent.
- (e) Construct an asymptotically valid feasible 95% confidence interval for θ .

2. Assume that $y_i^* \perp\!\!\!\perp t_i \mid \mathbf{x}_i$ (i.e., conditional independence, unconfoundedness).

- (a) Show that $\mathbb{E}[t_i \rho(y_i, \mathbf{x}_i; \theta)/p(\mathbf{x}_i)] = \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where $p(\mathbf{x}_i) = \mathbb{P}(t_i = 1 \mid \mathbf{x}_i) > 0$.
- (b) Assuming $\max_{1 \leq i \leq n} |\hat{p}_n(\mathbf{x}_i) - p(\mathbf{x}_i)| = o_{\mathbb{P}}(1)$ and $\min_{1 \leq i \leq n} p(\mathbf{x}_i) \geq p_{\min} > 0$, provide sufficient conditions for $M_n(\theta) \rightarrow_{\mathbb{P}} \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{\hat{p}_n(\mathbf{x}_i)}.$$

(c) The previous results motivate the estimator

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \frac{\rho(y_i, \mathbf{x}_i; \theta)}{\hat{p}_n(\mathbf{x}_i)},$$

which is consistent for θ_0 under regularity conditions on $\theta \mapsto \rho(\cdot, \cdot; \theta)$ and the (first-step) estimator $\hat{p}_n(\cdot)$. In particular, to simplify the problem, assume that $\hat{p}_n(\mathbf{x}) = p(\mathbf{x}; \hat{\gamma})$, where $\gamma \mapsto p(\cdot; \gamma)$ is a known smooth function, and the parametric estimator $\hat{\gamma}$ admits an asymptotic linear representation as follows:

$$\sqrt{n}(\hat{\gamma} - \gamma_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \vartheta_i + o_{\mathbb{P}}(1), \quad \text{where } \{\vartheta_i\}_{i=1}^n \text{ i.i.d. with } \mathbb{E}[\vartheta_i] = 0, \mathbb{V}[\vartheta_i] < \infty.$$

Provide sufficient conditions so that

$$\sqrt{n}(\hat{\theta}_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i + o_{\mathbb{P}}(1) \rightsquigarrow \text{Normal}(0, \Sigma_0),$$

where $\{\psi_i\}_{i=1}^n$ are mean-zero i.i.d., and identify the exact form of ψ_i and Σ_0 .

(d) Propose a consistent estimator of Σ_0 .

(e) Construct an asymptotically valid 95% confidence interval for θ_0 .

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$